Project Progress Report: Modelling the Spread of Information Through Instagram Reposts

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1 Introduction

Social media platforms such as Instagram have significantly changed the way information spreads in modern society. Posts, videos, and trends can rapidly gain popularity through reposting, sharing, and algorithmic amplification. When users repeatedly encounter the same content through reposts, they may eventually decide to share it with their own followers. This creates cascades of information spread similar to epidemics in biological systems.

Understanding this process is vital for predicting viral trends, managing public opinion, and analyzing marketing campaigns. Traditional models often assume a fixed exposure threshold; however, real user behavior is heterogeneous. This project proposes a mathematical model that allows sharing or loss of interest to occur at any exposure level, capturing the complex dynamics of modern digital interaction.

2 Literature Review

Mathematical modelling of information diffusion is often inspired by epidemiological frameworks. One of the earliest foundational works is the rumor model by Daley and Kendall (1965), which divides individuals into ignorants, spreaders, and stiflers. Recent research has extended these ideas to network structures, focusing on influence maximization (Kempe et al., 2003).

Experimental studies, such as those by Centola (2010), demonstrate that complex behaviors often require reinforcement through multiple exposures. Furthermore, Rogers' diffusion framework (2003) highlights that individuals—ranging from early adopters to the laggards—respond differently to repeated exposure. Our model builds upon these theories by introducing multiple exposure classes (I_i) to represent varying probabilities of engagement.

3 Model Formulation and Justification

The population N is divided into three main categories: $I_i(t)$ (users exposed i times), $S(t)$ (active sharers), and $R(t)$ (saturated users). We model the exposure rate as $\lambda(t) = \beta\frac{S(t)}{N}$,

where β represents reposting intensity.

3.1 Model Equations

The system of ordinary differential equations (ODEs) is defined as:

Exposure Classes:

$$\frac{dI_0}{dt} = -\lambda(t)I_0 \tag{1}$$

$$\frac{dI_i}{dt} = (1 - \rho_{i-1} - \sigma_{i-1})\lambda(t)I_{i-1} - \lambda(t)I_i \quad \text{for } i = 1, \ldots, m-1 \tag{2}$$

$$\frac{dI_m}{dt} = (1 - \rho_{m-1} - \sigma_{m-1})\lambda(t)I_{m-1} - (\rho_m + \sigma_m)\lambda(t)I_m \tag{3}$$

Sharers and Saturated Users:

$$\frac{dS}{dt} = \sum_{i=0}^{m} \rho_i \lambda(t) I_i - \gamma S \tag{4}$$

$$\frac{dR}{dt} = \sum_{i=0}^{m} \sigma_i \lambda(t) I_i + \gamma S \tag{5}$$

3.2 Table 1: Model Variables and Parameters

Symbol	Description
$I_i(t)$	Users exposed i times but not sharing
$S(t)$	Active sharers
$R(t)$	Saturated users (stopped engagement)
β	Reposting intensity parameter
γ	Rate at which sharers stop sharing (decay)
ρ_i	Probability of sharing at exposure level i
σ_i	Probability of losing interest at level i

4 Preliminary Results

Current analysis focuses on the basic reproduction number (R_0), which determines if information will go viral. Preliminary results suggest that increasing β and the early exposure sharing probabilities (ρ_0, ρ_1) significantly increases the peak number of sharers (S_{max}). Conversely, a high decay rate γ effectively limits the total reach of the content.

5 Future Work

Future steps include:

1. Performing stability analysis on the equilibrium points.

2. Implementing numerical simulations in Python/MATLAB to visualize the timeline of infection.

3. Analyzing three specific case studies: Viral Memes, News Stories, and Awareness Campaigns.

6 References

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